



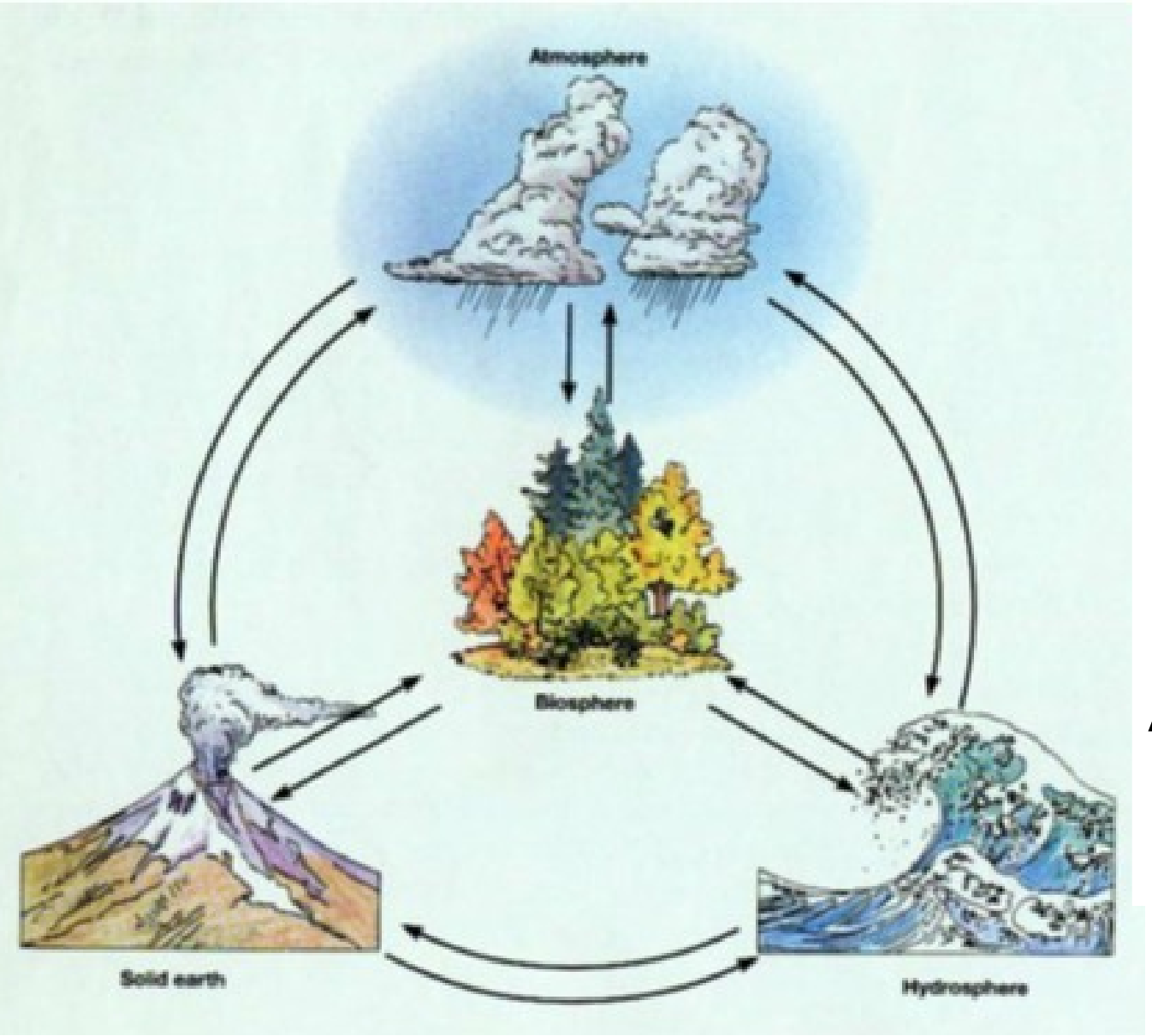


ESYS 104

Fall 2026

The Earth System

The Earth System



Lithosphere
- Crust
- Deep Earth

Biosphere
- Humans

Hydrosphere
- Cryosphere

Atmosphere

Deep Time

Instructors

Professor: William Minarik
william.minarik@mcgill.ca
Office: FDA (Adams Building) Room 215

Teaching Assistants: Yan Su, Jordan Thakar, plus two more

Syllabus I

Course Topics:

No.	*Date*	*Subject*	
1.	31-Aug	Introduction	
2.	02-Sep	Solar system/Earth materials	
3.	07-Sep	**Labour Day**	
4.	09-Sep	Plate tectonics	
5.	14-Sep	Reservoirs and resources	
6.	16-Sep	Energy and the atmosphere	Quiz #1
7.	21-Sep	Atmospheric circulation	
8.	23-Sep	Ocean circulation/Arctic Ocean	
9.	28-Sep	Hydrology, soils and their productivity	Assig #1 due
10.	30-Sep	Biomes, ecology and ecosystems	Quiz #2
	05-Oct	**Classes cancelled** -Quebec elections	
11.	07-Oct	Ecosystem resilience and regime shifts	Feedbacks question
	12-Oct	**Fall Break**	
	14-Oct	**Fall Break**	
12.	19-Oct	Deep time/Earth history	Assig #2 due
13.	21 Oct	Himalayas, weathering	
14.	26-Oct	**MIDTERM exam** (in class)	
15.	28-Oct	Snowball Earth (evolution of atmosphere)	

Draft -Subject to change

Syllabus II

No.	*Date*	*Subject*	
16.	02-Nov	Glacial Cycles	Quiz #3
17.	04-Nov	Global Climate Change & PaleoClimate	Proxy question
18.	09-Nov	Arctic/Antarctic Climate Change	
19.	11-Nov	Global population growth	
20.	16-Nov	The Sahel and the subtropics	Assig #3 due
21.	18-Nov	Coastal Processes	
22.	23-Nov	Earth System hazards	Quiz #4
23.	25-Nov	Fossil Fuels	
24.	30-Nov	Energy alternatives	Assig #4 due
25.	02-Dec	Planetary Boundaries	Quiz #5
26.	03-Dec	Tipping Points in the Climate System; Review	

Draft -Subject to change

Evaluation

Marking:
40% Four assignments, spaced ~every two weeks
20% Midterm Exam (in class)
25% Final Exam

15% quizzes and short writings plus in-class exercises

The lowest score for these quizzes, etc. will be dropped, so you can miss one without issue.

Additional Information

"McGill University values **academic integrity**. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the Code of Student Conduct and Disciplinary Procedures (see www.mcgill.ca/students/srr/honest/ for more information)."

"In accord with McGill University's Charter of Students' Rights, students in this course have the right to submit in **English** or in **French** any written work that is to be graded."

In the event of extraordinary circumstances beyond the University's control, the evaluation scheme in a Course is subject to change, provided that there be timely communications to the students regarding the change.

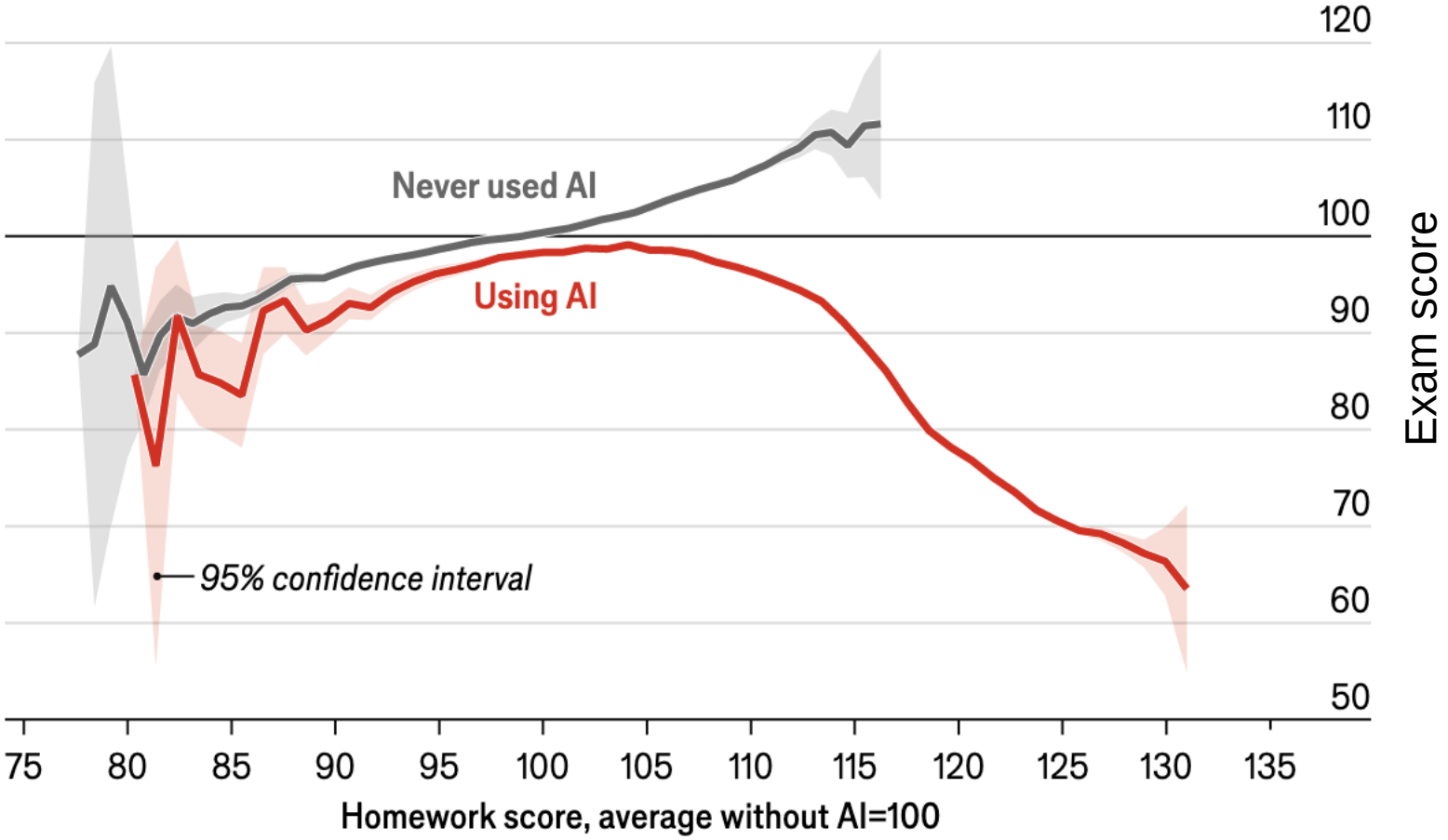
Use of LLMs (ChatGPT, Gemini, Copilot, AISlop etc.) is probably unavoidable. However, if the LLM answer is different from the class material, the class material will be the correct answer. Using LLMs also means you're not thinking the problem through, which will disadvantage you on the exams.

Use of Smartglasses (Meta glasses, etc.) in class is strictly prohibited.

Thinking about it

China*, average exam score

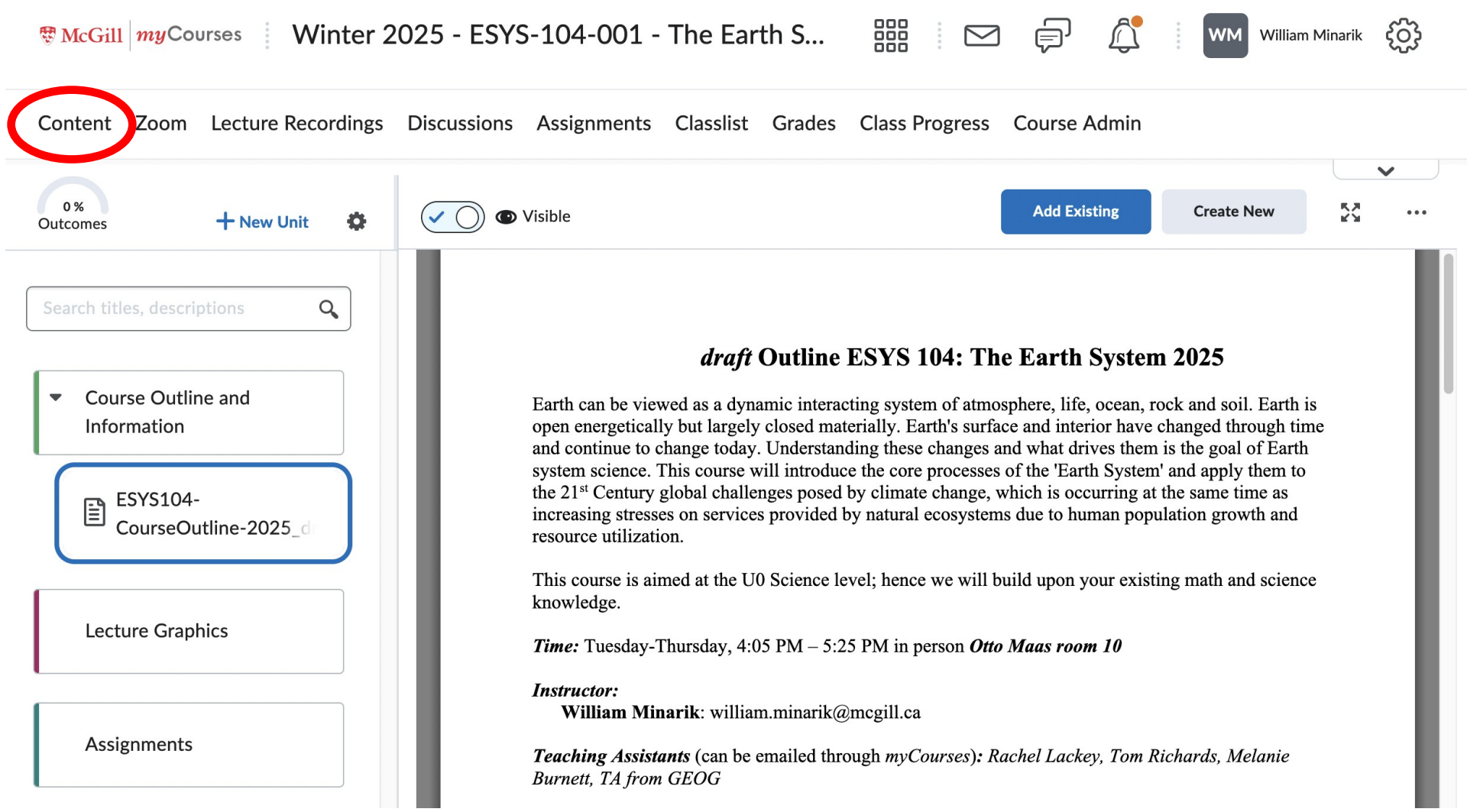
January 2023-June 2025, average without AI=100



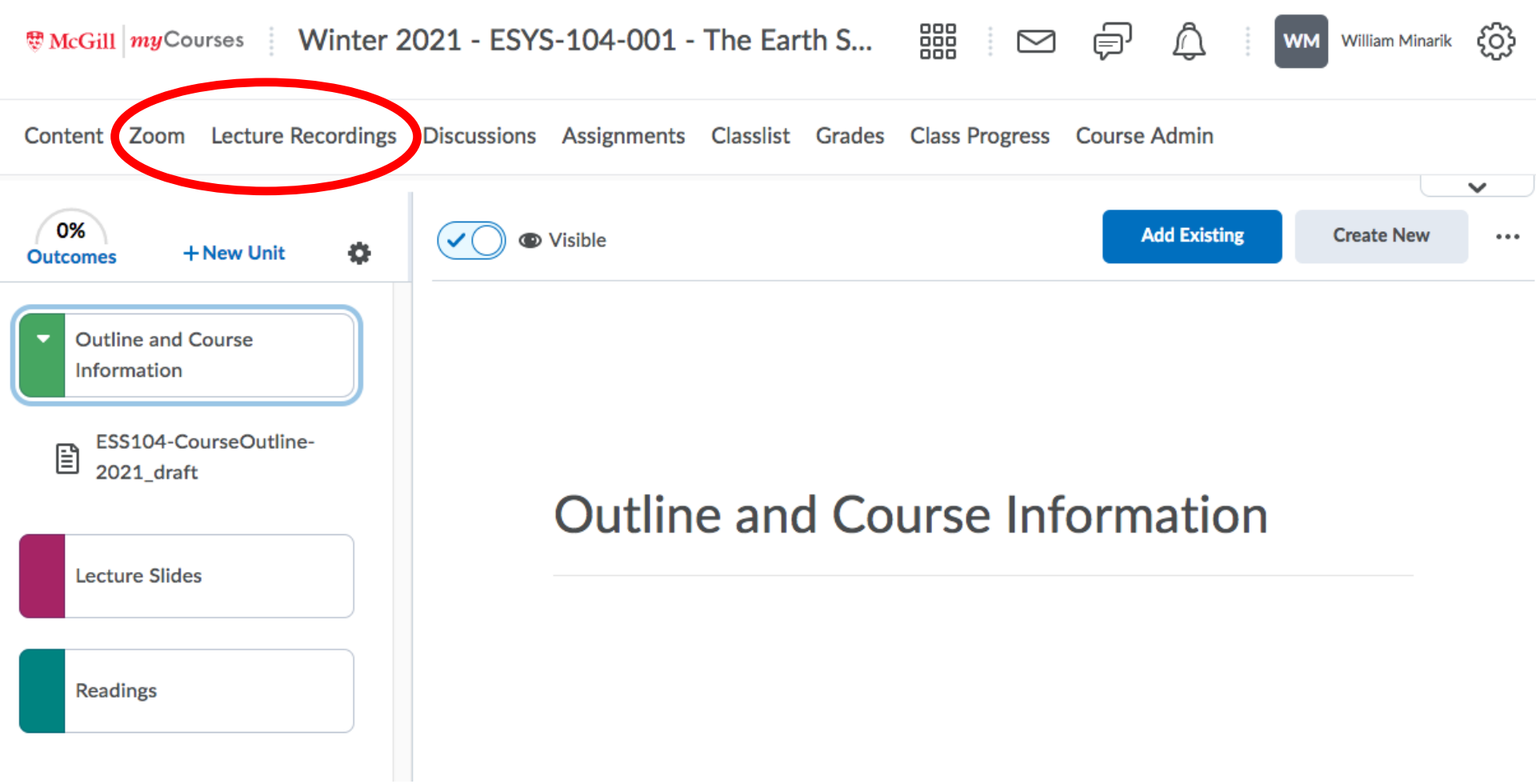
*26,811 students aged 12-18

Source: "The generative AI learning penalty: evidence from Chinese secondary education", by D. Strömberg et al., preprint June 2026

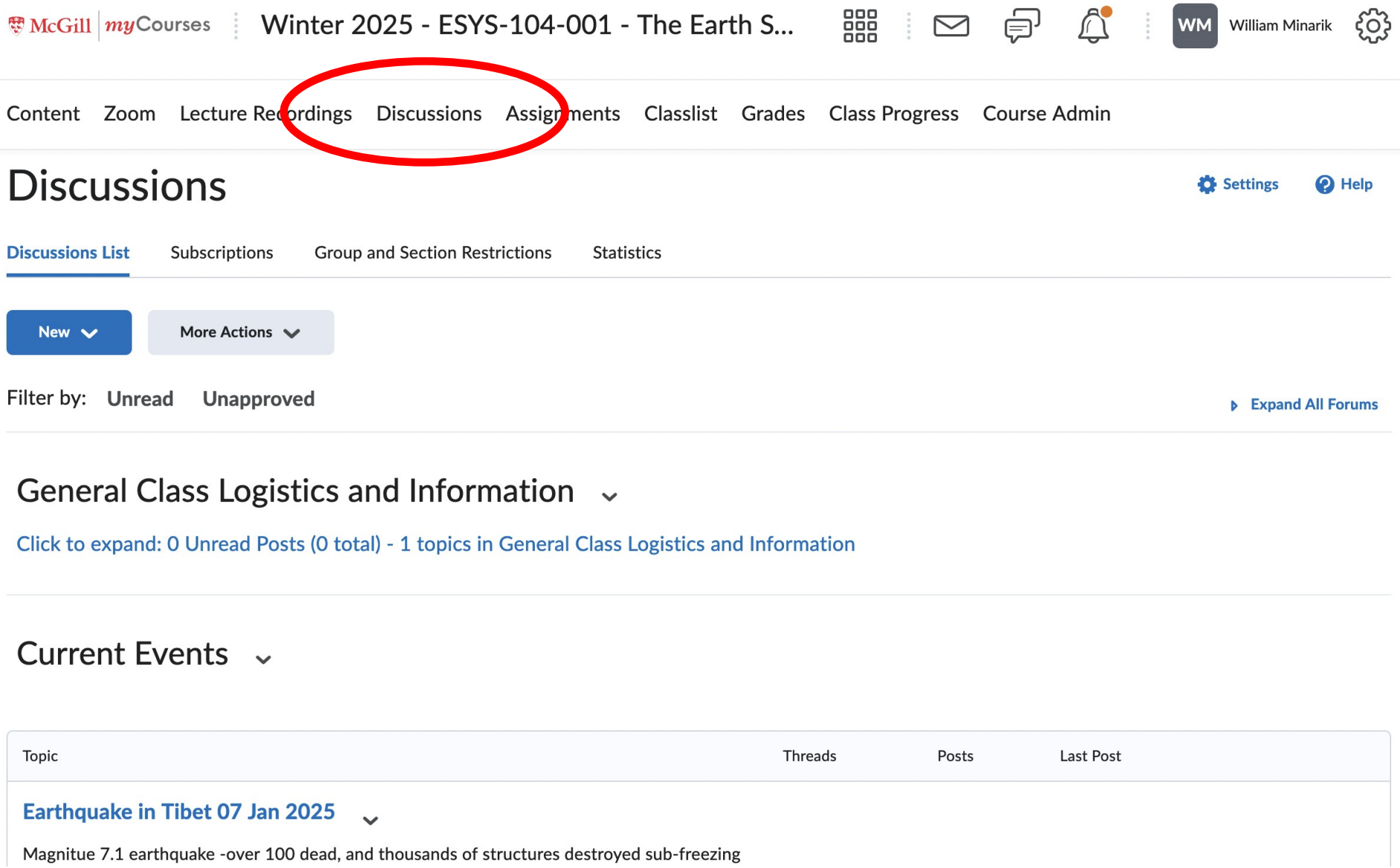
MyCourses


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Winter 2025 - ESYS-104-001 - The Earth S...
WM
William Minarik
Content
Zoom
Lecture Recordings
Discussions
Assignments
Classlist
Grades
Class Progress
Course Admin
0 %
Outcomes
New Unit
Visible
Add Existing
Create New
Search titles, descriptions
Course Outline and Information
ESYS104-CourseOutline-2025_d
Lecture Graphics
Assignments
draft Outline ESYS 104: The Earth System 2025
Earth can be viewed as a dynamic interacting system of atmosphere, life, ocean, rock and soil. Earth is open energetically but largely closed materially. Earth's surface and interior have changed through time and continue to change today. Understanding these changes and what drives them is the goal of Earth system science. This course will introduce the core processes of the 'Earth System' and apply them to the 21st Century global challenges posed by climate change, which is occurring at the same time as increasing stresses on services provided by natural ecosystems due to human population growth and resource utilization.
This course is aimed at the U0 Science level; hence we will build upon your existing math and science knowledge.
Time: Tuesday-Thursday, 4:05 PM – 5:25 PM in person Otto Maas room 10
Instructor:
William Minarik: william.minarik@mcgill.ca
Teaching Assistants (can be emailed through myCourses): Rachel Lackey, Tom Richards, Melanie Burnett, TA from GEOG

MyCourses


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Lecture Recordings
Discussions
Assignments
Classlist
Grades
Class Progress
Course Admin
0%
Outcomes
+ New Unit
Visible
Add Existing
Create New
Outline and Course Information
ESS104-CourseOutline-2021_draft
Lecture Slides
Readings
Outline and Course Information

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Discussions
Settings
Help
Discussions List
Subscriptions
Group and Section Restrictions
Statistics
New
More Actions
Filter by:
Unread
Unapproved
Expand All Forums
General Class Logistics and Information
Click to expand: 0 Unread Posts (0 total) - 1 topics in General Class Logistics and Information
Current Events
Topic
Threads
Posts
Last Post
Earthquake in Tibet 07 Jan 2025
Magnitue 7.1 earthquake -over 100 dead, and thousands of structures destroyed sub-freezing

Reading Materials

Lecture 2: Earth Materials (Wednesday, 2 Sep)
https://wiki.seg.org/wiki/Layers_of_the_Earth

https://opentextbc.ca/physicalgeology2ed/chapter/1-4-minerals-and-rocks/

https://opentextbc.ca/physicalgeology2ed/chapter/3-1-the-rock-cycle/

Look over: Chapter 2 Minerals

Lecture 3: Plate Tectonics (Wednesday, 9 Sep)
https://opentextbc.ca/physicalgeology2ed/chapter/1-5-fundamentals-of-plate-tectonics/

Look over: Chapter 10 Plate Tectonics

Physical Geology – 2nd Edition
By Steven Earle
https://opentextbc.ca/physicalgeology2ed/


THE FACULTY OF SCIENCE PRESENTS – THE 42ND EDITION OF
SOUP AND SCIENCE
SEPTEMBER 14 – 18, 2026
THOMSON HOUSE – BALLROOM
11:45 – 12:45
LUNCH PROVIDED
FIRST COME, FIRST SERVED
N_2O
CH_4
CO_2
A G C T
10011001
11010100
00010011
$E=mc^2$
$PV=nRT$
Ψ
$\pm$
$=$

The Earth System: ESYS 104
ATOC-EPSC-GEOG



Harrison Schmidt, Apollo 17

How to these portions of the system interact?

Canadian High Arctic



Ellesmere Island today



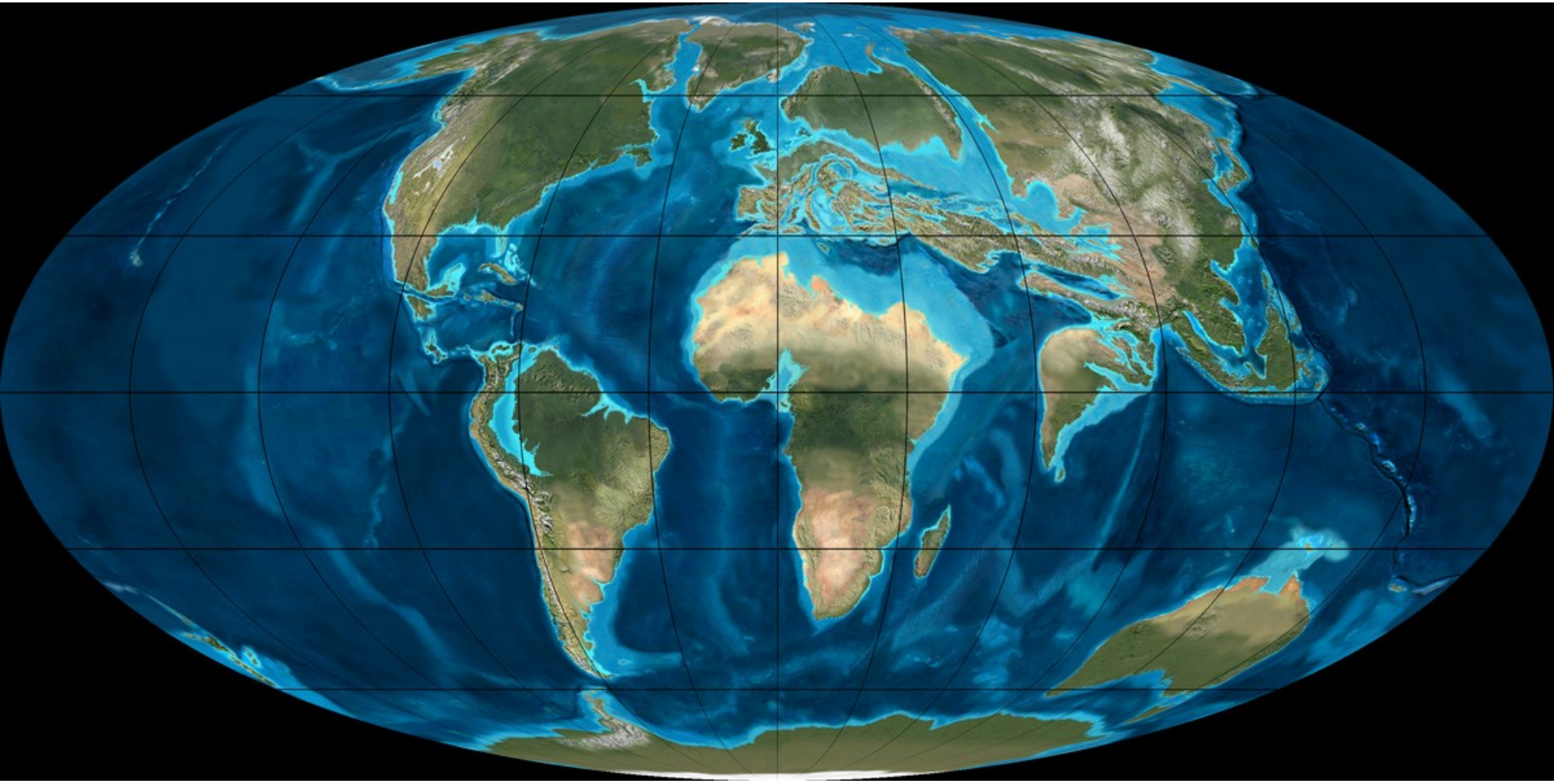
Hippos on Ellesmere Island!



Coryphodon

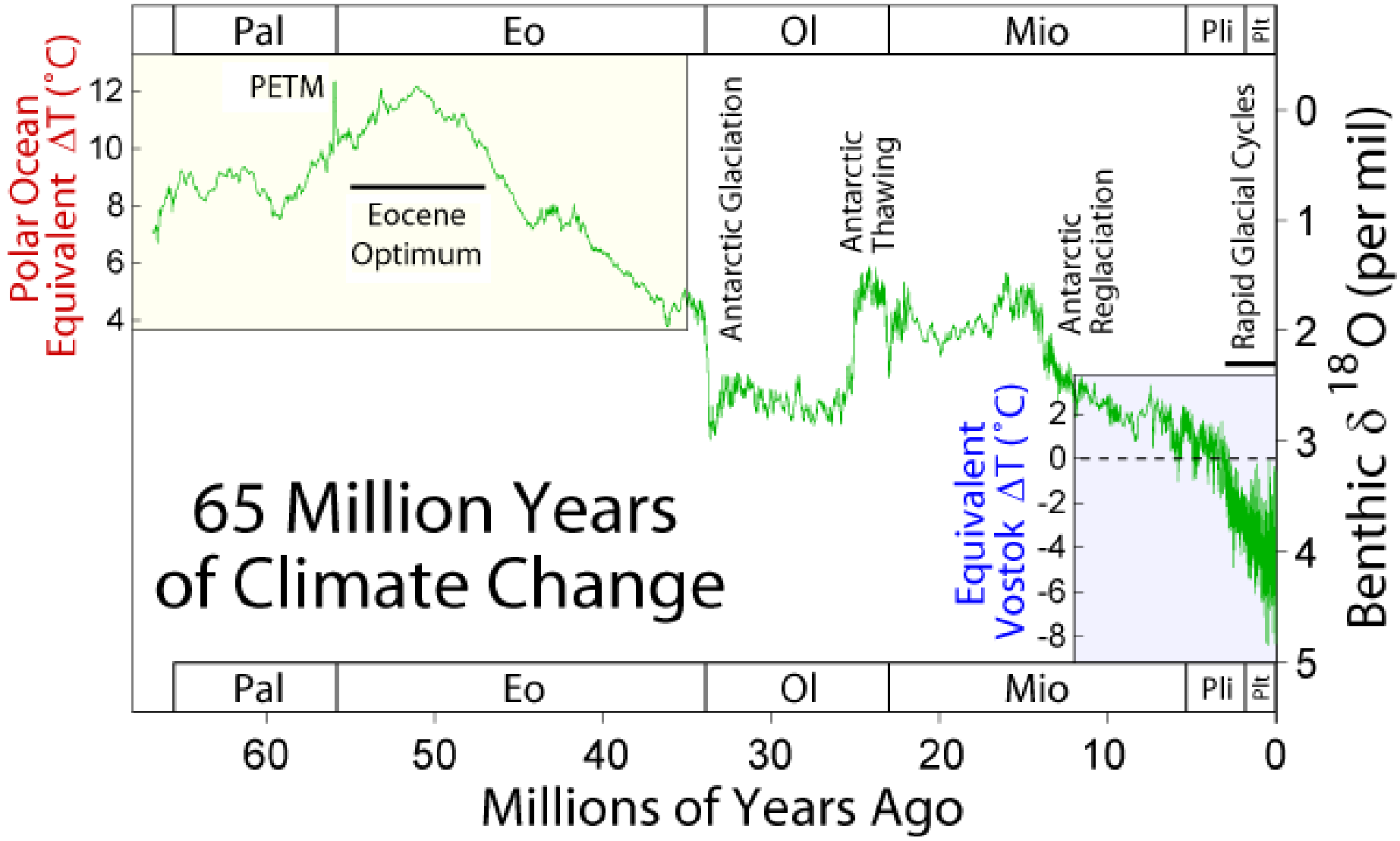
55 Million Years ago

50 Ma Geography



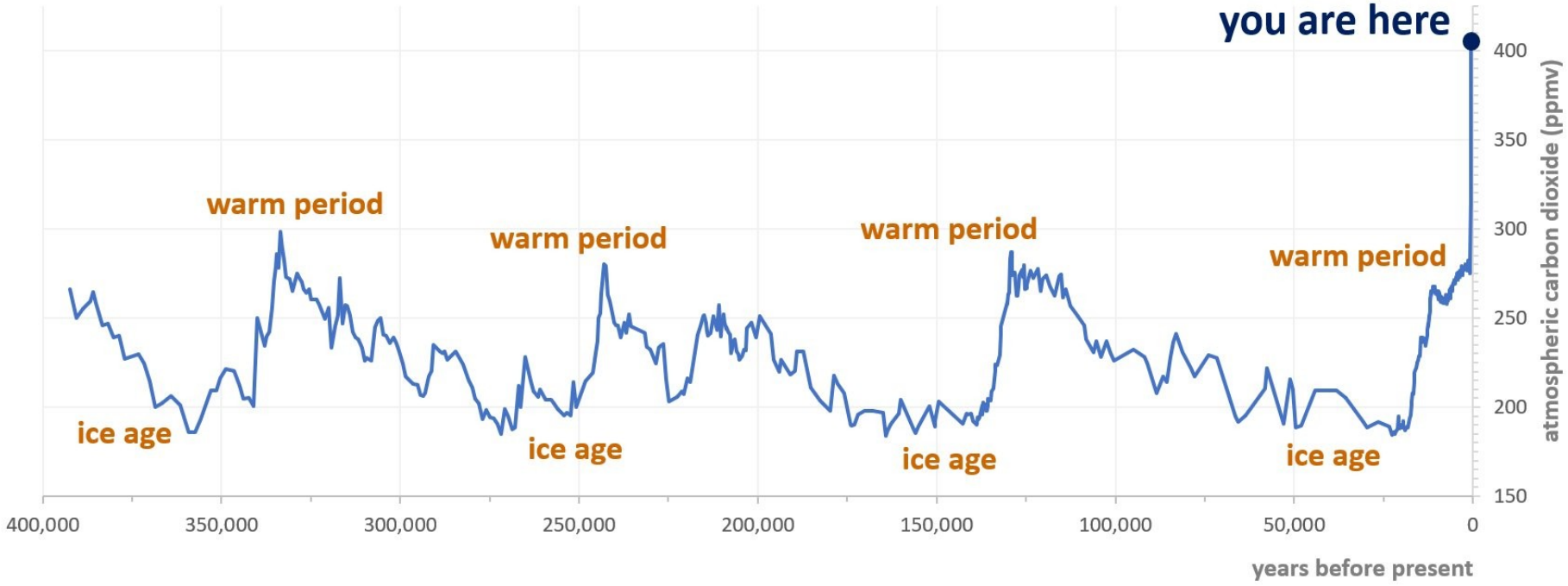
A similar arrangement of continents -plate tectonics is slow

Climate change during the age of Mammals



Robert Rohde

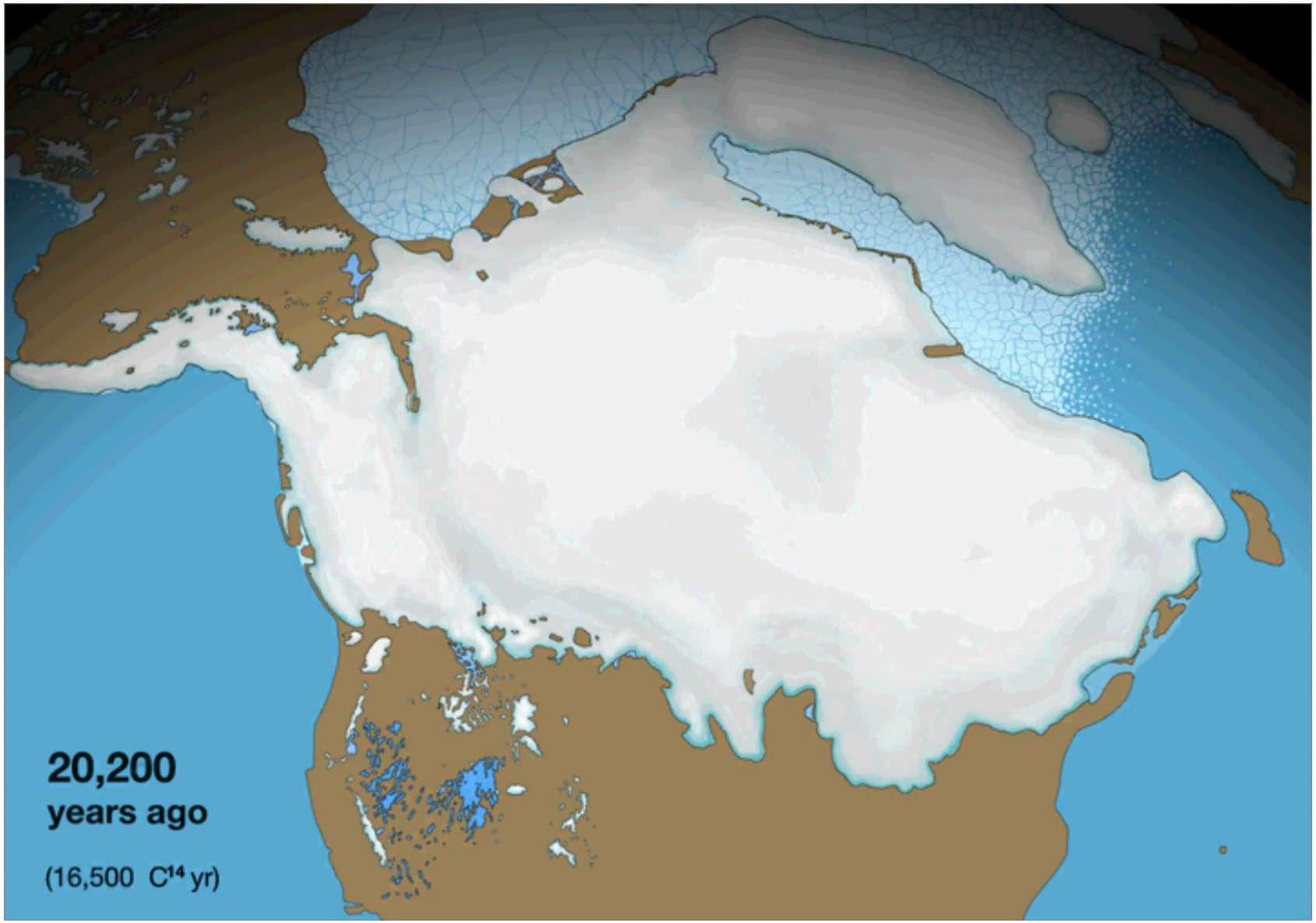
Deep Time CO_2 variations



Vostok, Antarctic Ice Core

Low CO_2 concentrations represent times of Continental Glaciation

Dramatic Past Climate Change



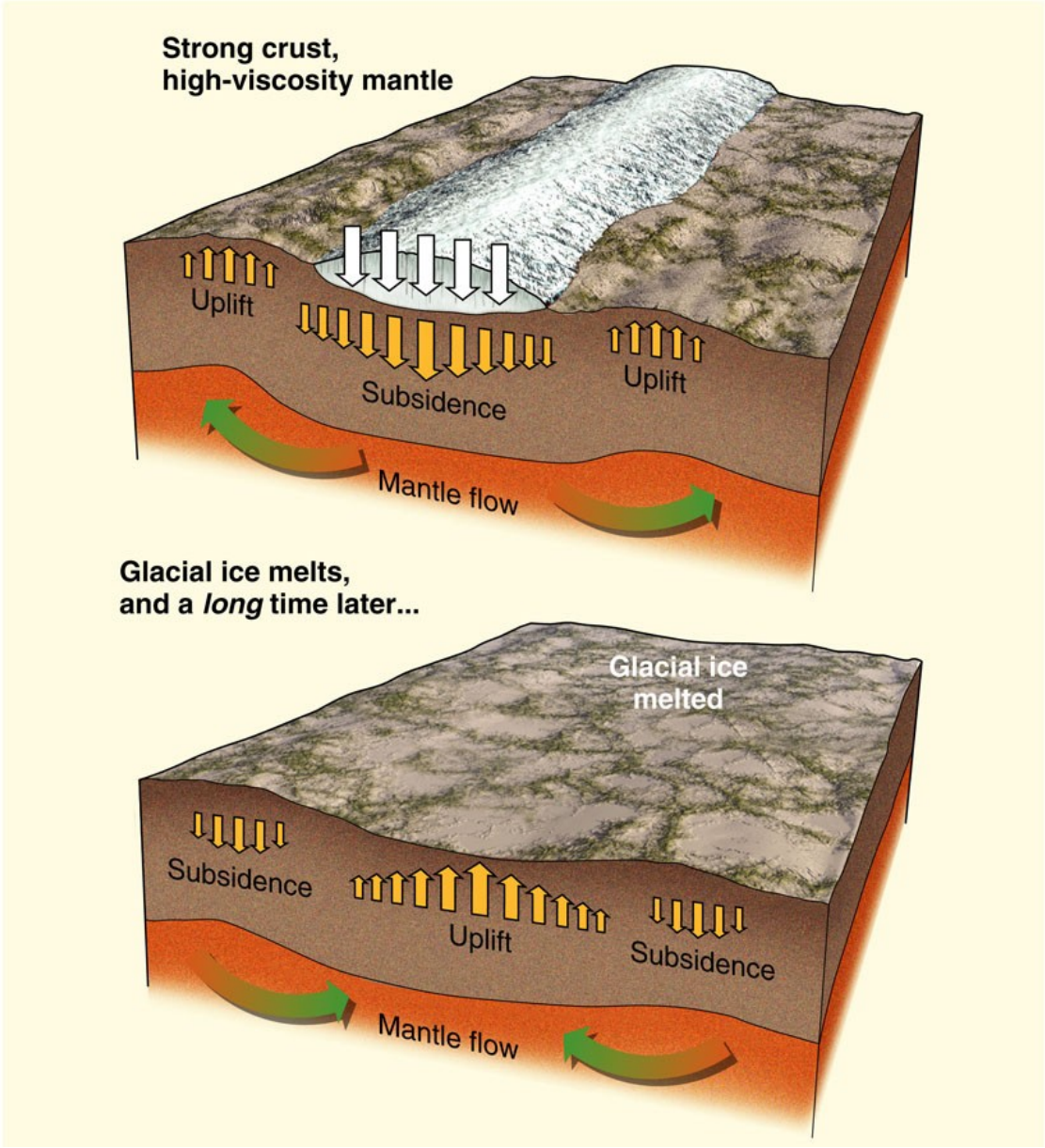
Ice extent -last glacial maximum

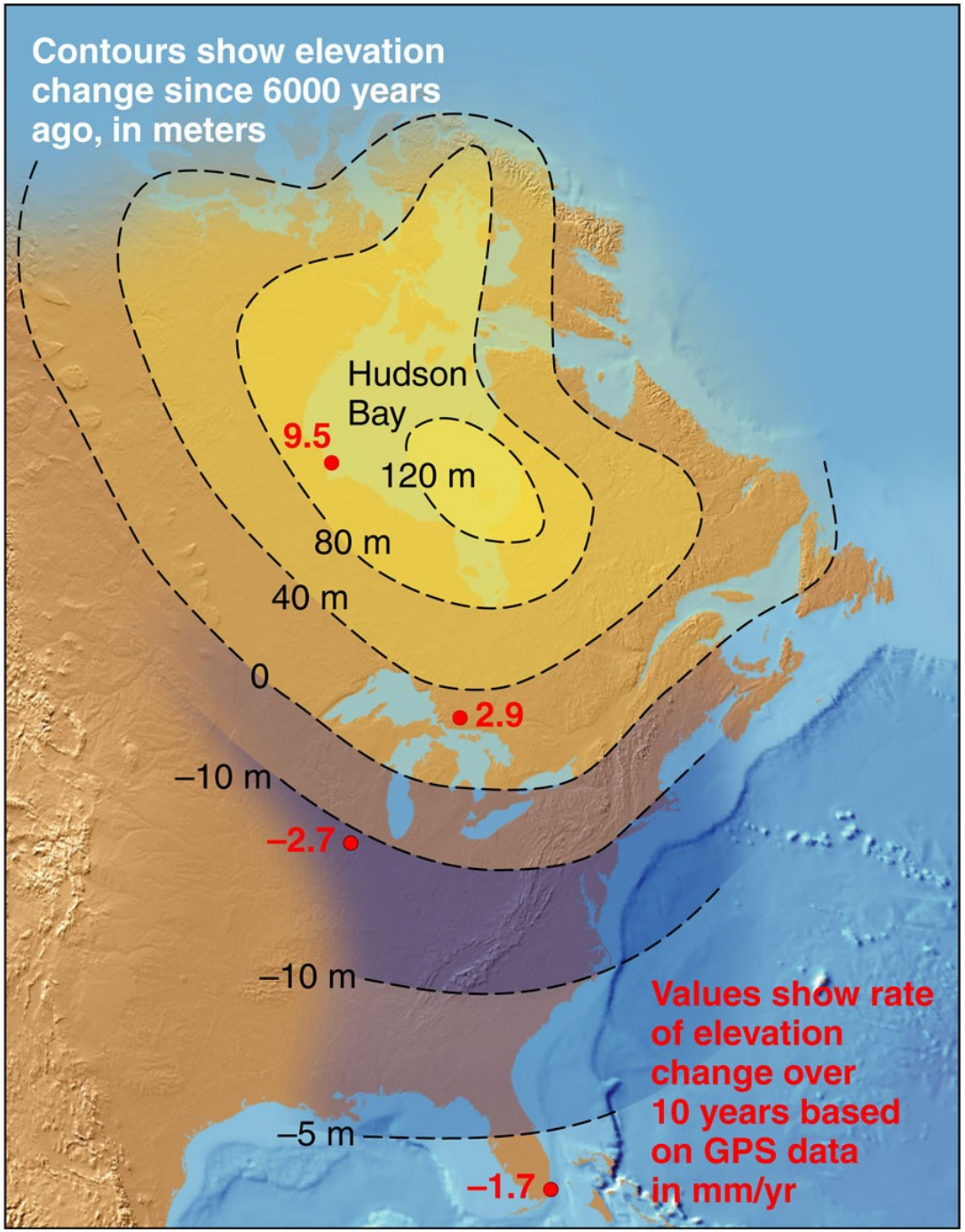
Beach terraces -Hudson's Bay



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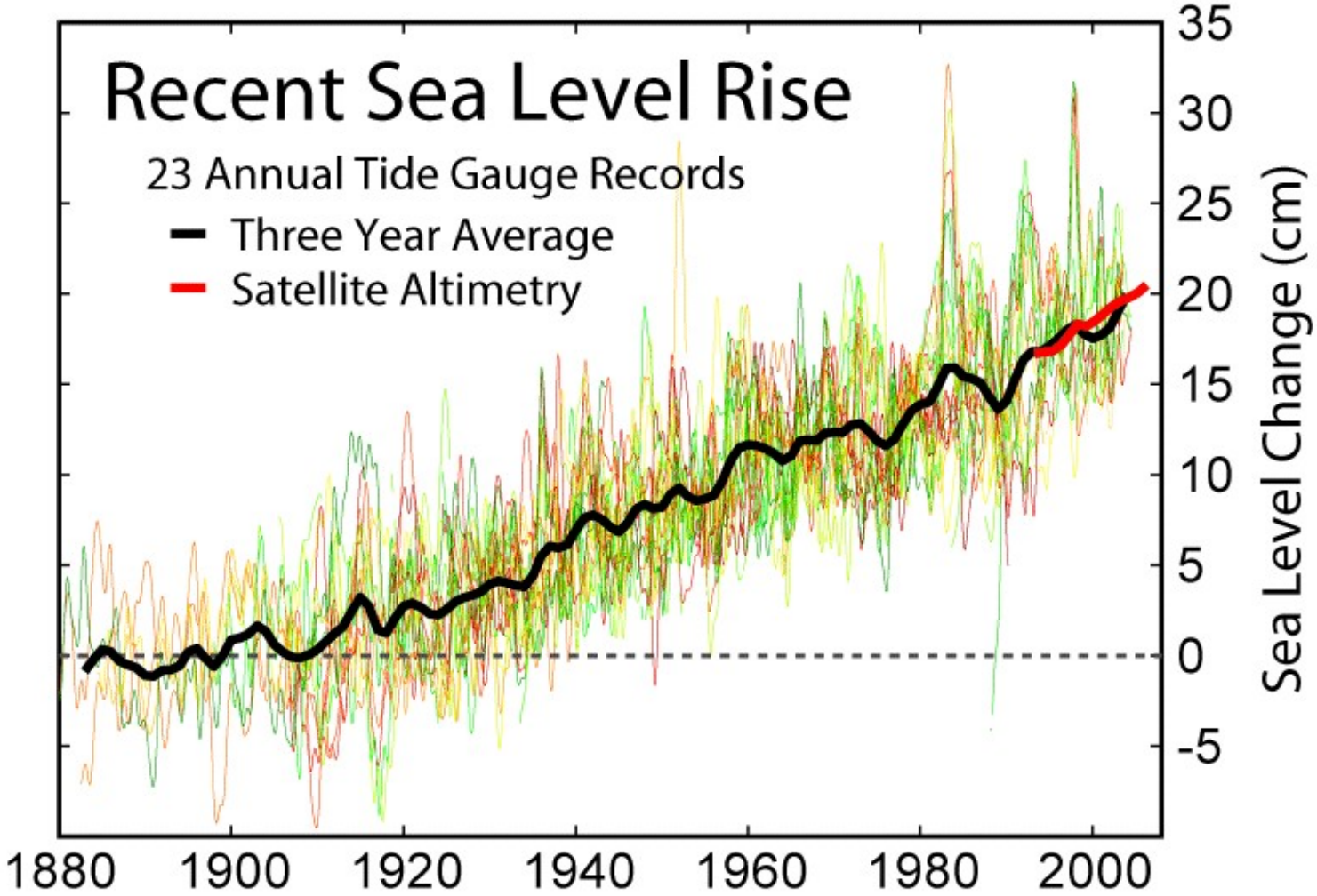
Isostasy and Glacial Rebound



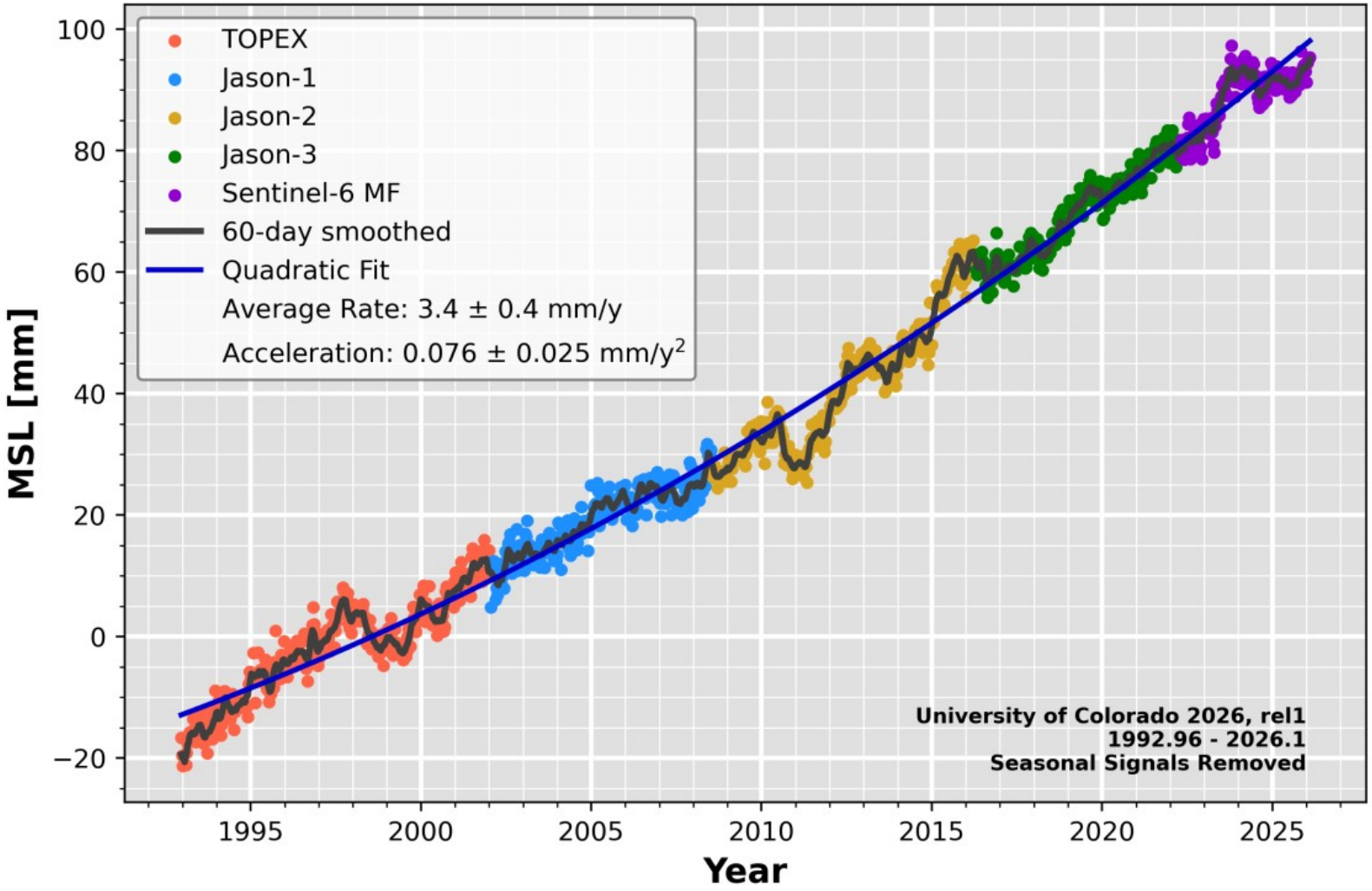


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Current global average Sea Level Rise, about 3.5 cm rise a decade, but accelerating

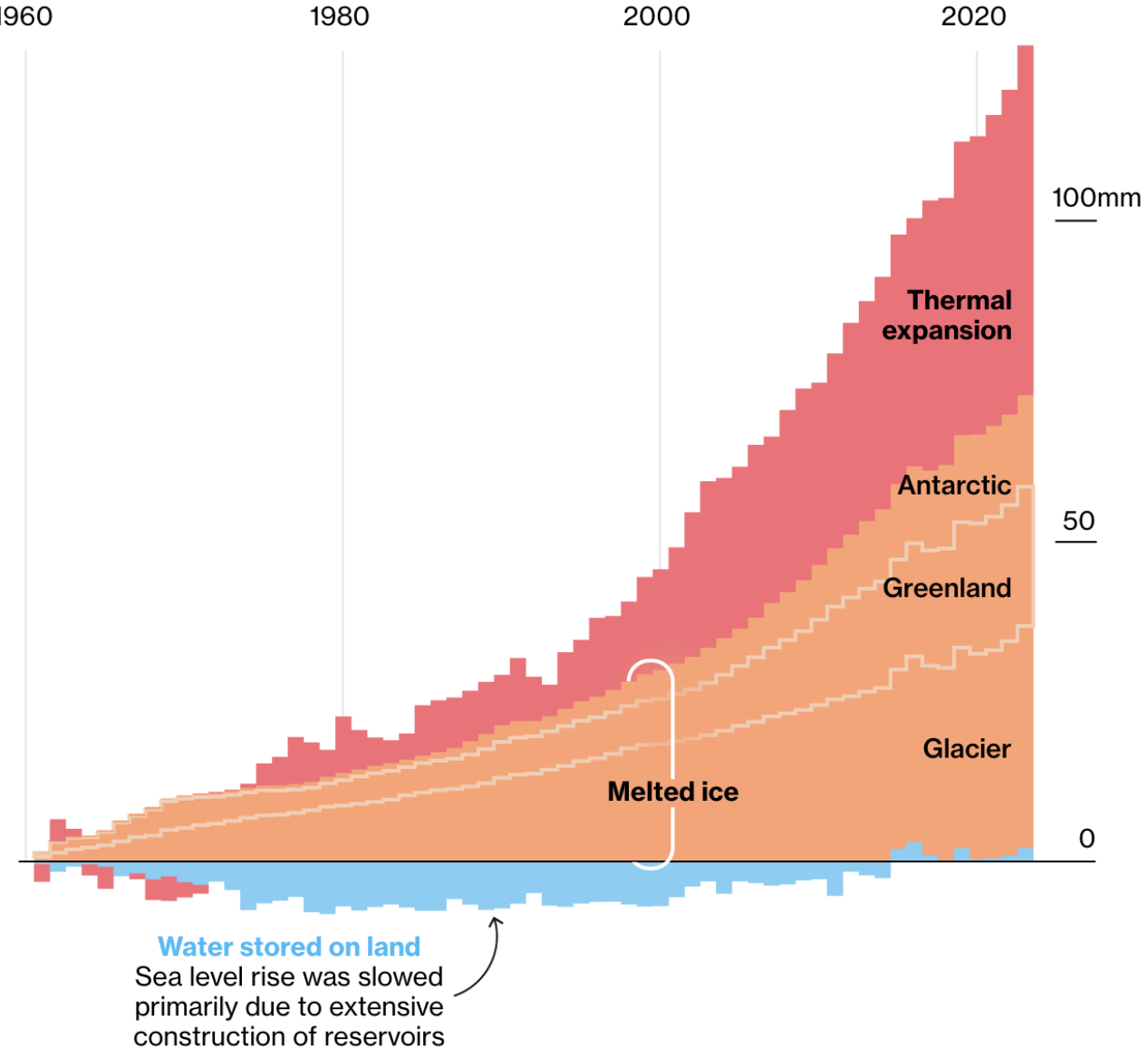


Current global average Sea Level Rise, about 3.5 cm rise a decade, but accelerating



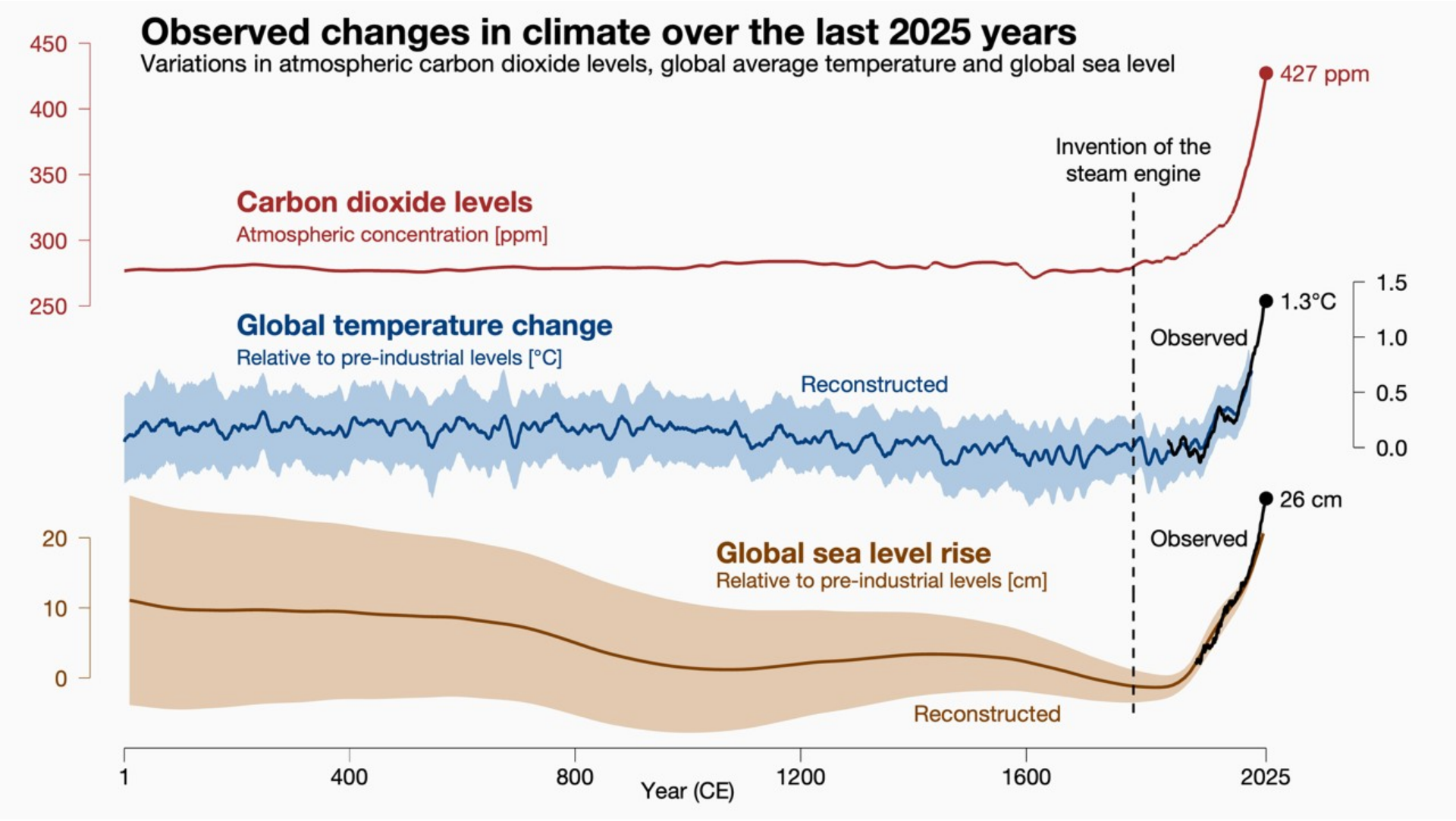
Melted Ice Makes Up More Than Half of Sea Level Increase

Contribution to sea level rise in millimeters

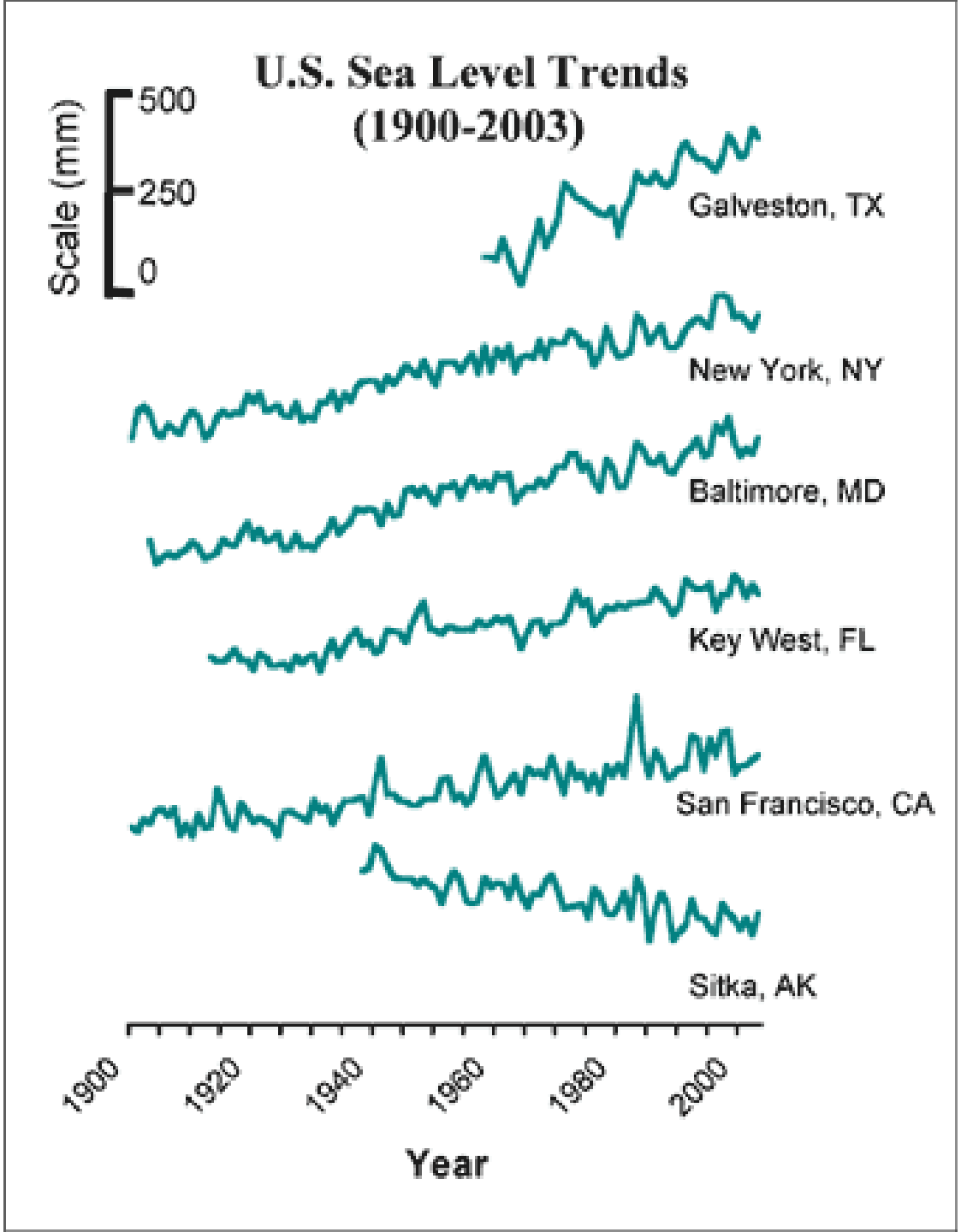


Source: Zheng et al. (2026). *Improved closure of the global mean sea level budget from observational advances since 1960*, Science Advances.

Observed (measured) changes

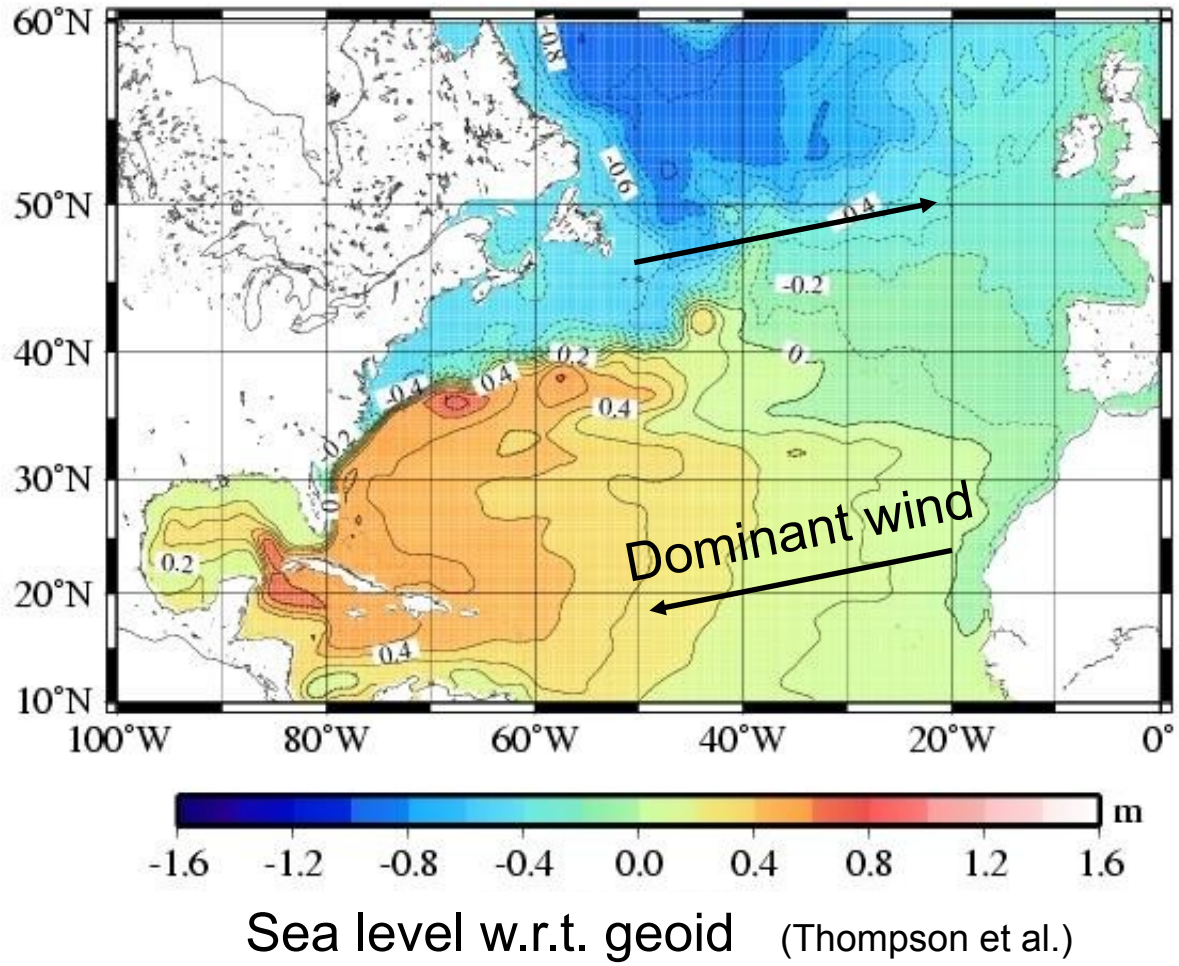


Proxy Data


U.S. Sea Level Trends
(1900-2003)
Scale (mm)
500
250
0
Galveston, TX
New York, NY
Baltimore, MD
Key West, FL
San Francisco, CA
Sitka, AK
1900
1920
1940
1960
1980
2000
Year

U.S.G.S.

Winds drag water, shaping mean sea level



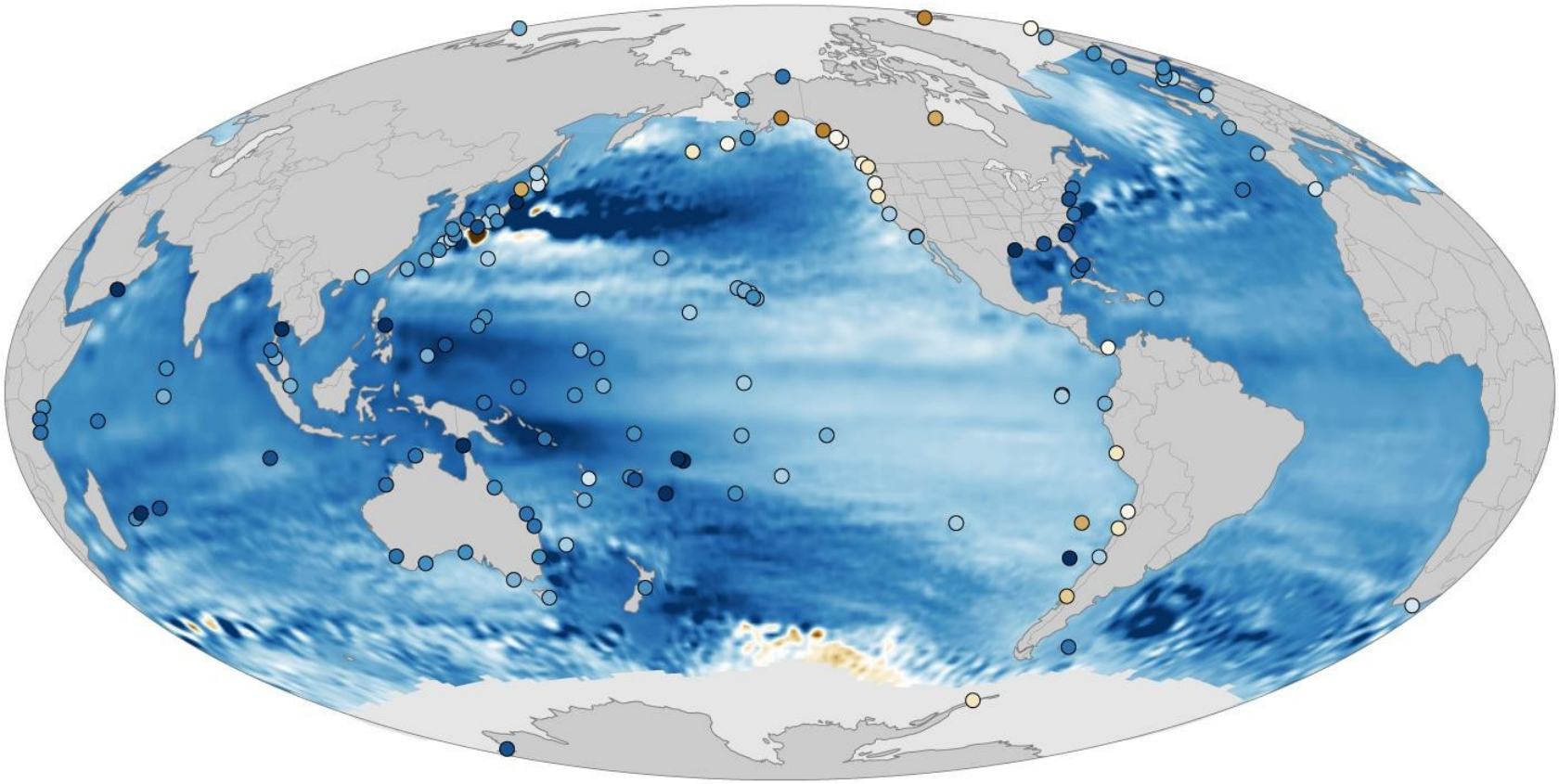
Sea level w.r.t. geoid (Thompson et al.)

Regional sea level may change with a change in atmospheric circulation.

…and ocean currents would change, changing sea temperatures.

Since temperature differences drive the atmosphere, a feedback will occur.

SEA LEVEL CHANGE (1993-2025)



1993–2025

Change in sea level (cm)

-20 0 20

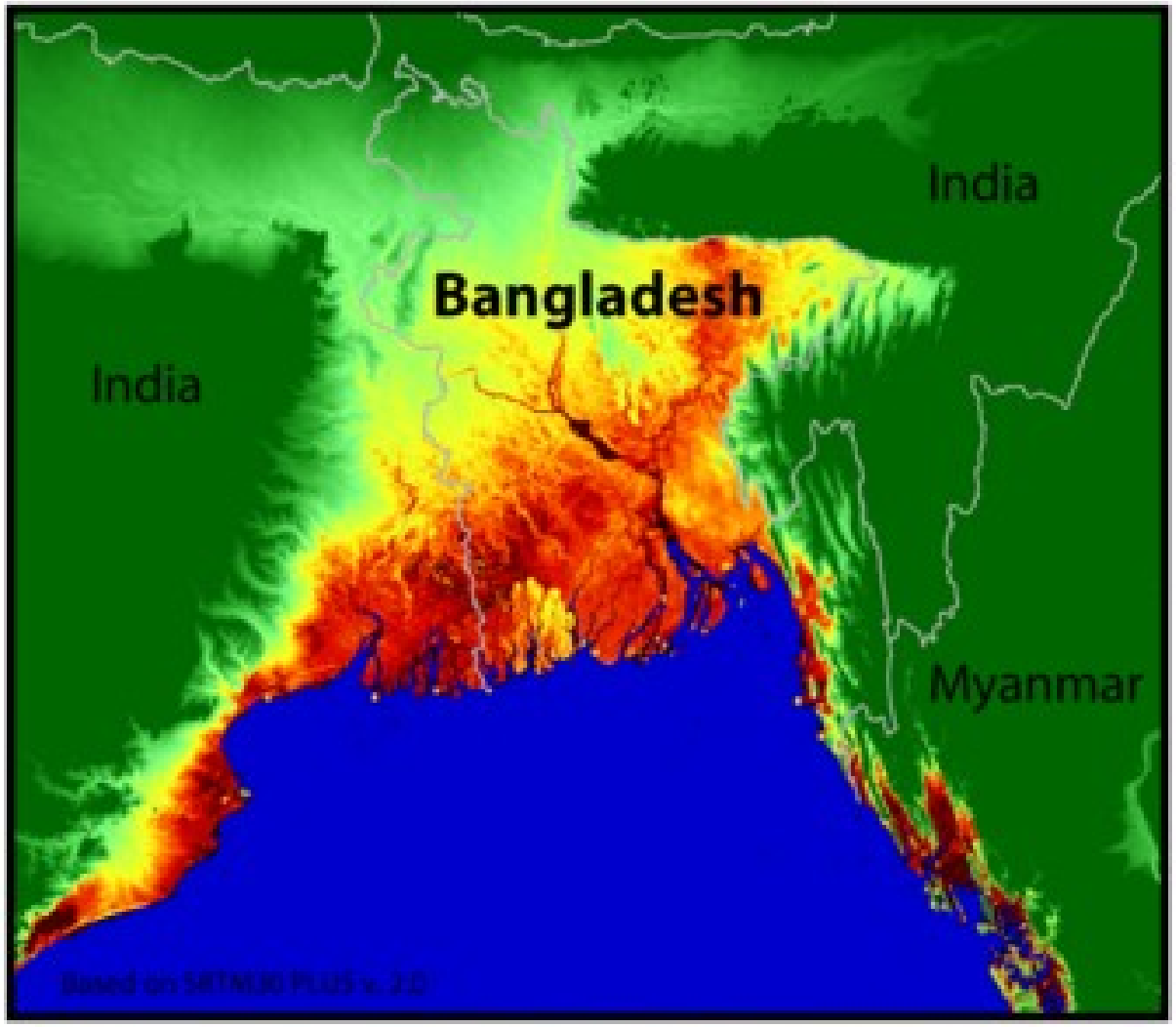
Climate.us
Data: UHSLC

Sea Level Rise

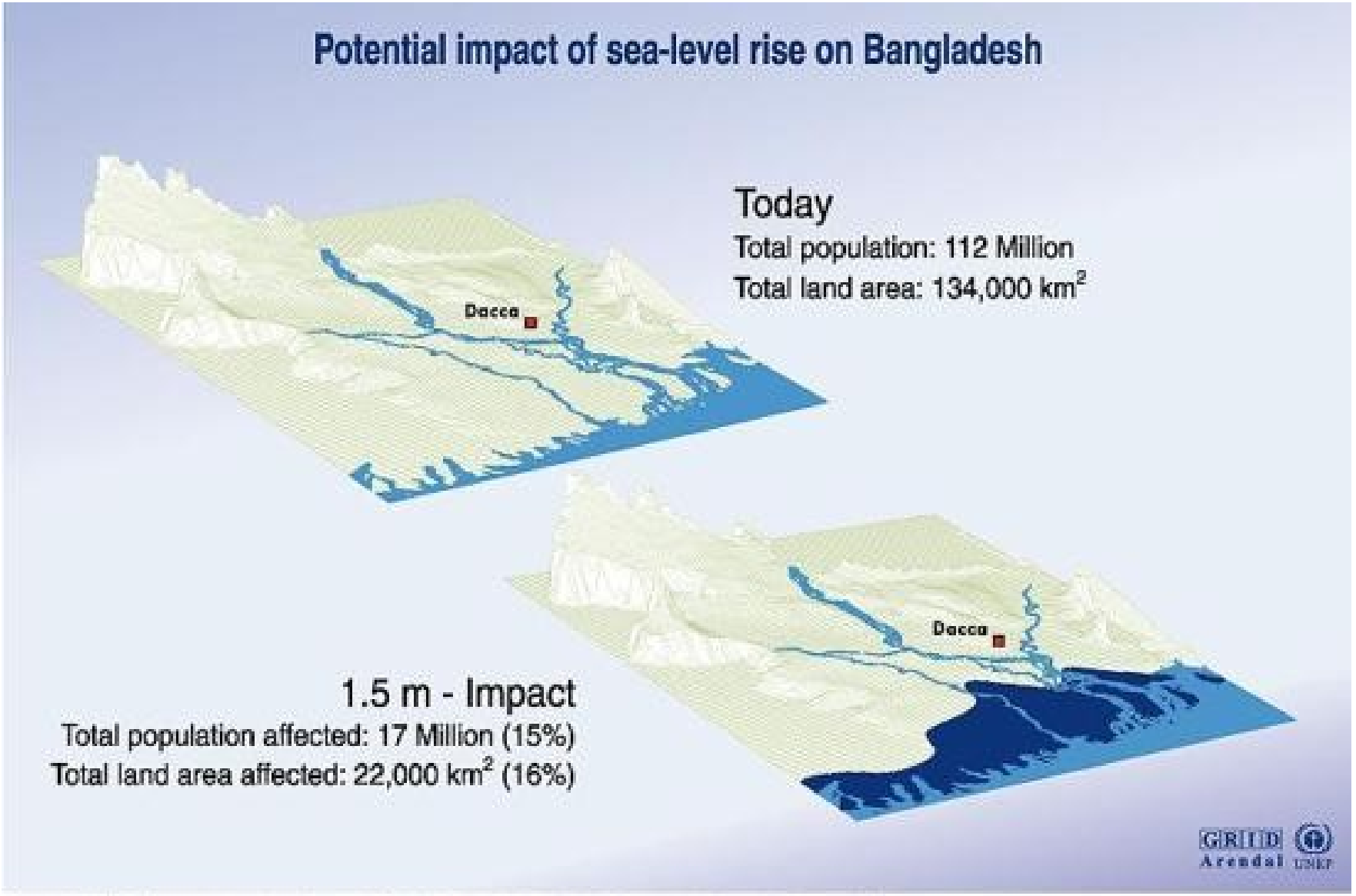


People tend to live along the coast (here, Bangladesh)

Sea Level Risks - Bangladesh



A 1.5 m rise in sea level will drown lands occupied by almost 20 million people



Source : UNEP/GRID Geneva; University of Dacca; JRO Munich; The World Bank; World Resources Institute, Washington D.C.

Miami Beach…



Discussion

Questions?

Particular Topics of Interest?

Math is the Language of Science

A sampling of math:

1) The residence time is the average time a substance spends in a reservoir. It's defined as the size of the reservoir divided by the flux of material in or out.

If your bathtub holds 100 liters of water, and the faucet is set to replace water at just the rate water leaves via the leaky drain at 2 liters per minute, what is the residence time of water in the bath?

2) If the equation for a line is y = m*x + b (where m is the slope and b is the y-intercept), can you solve for x, if you are given y?

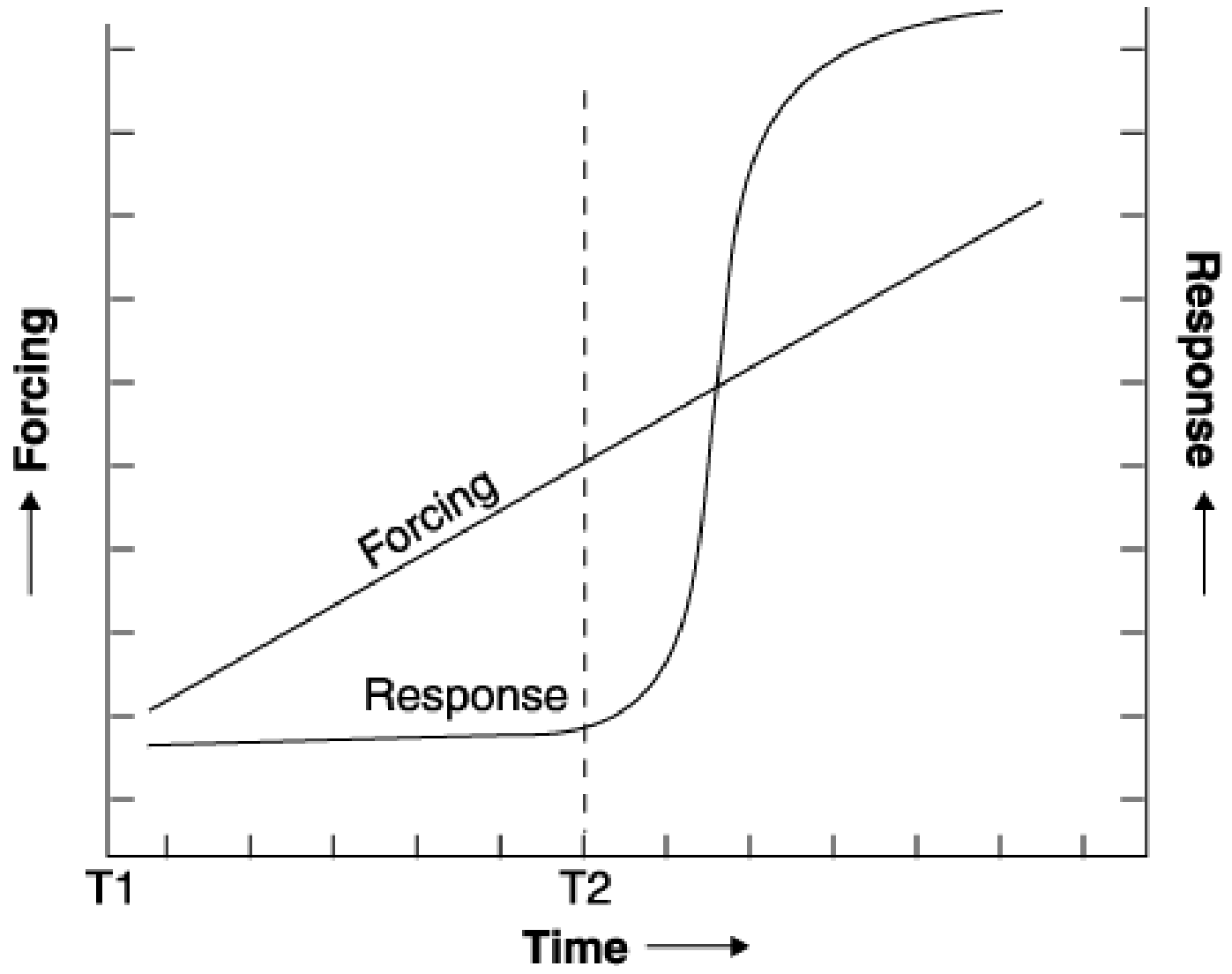
A sampling of math:

3) The half-life of a radioactive isotope is the time it takes for half of it to decay away.

If the half-life of tritium (radioactive hydrogen) is about 13 years, and a substance has a quarter of the tritium it started with, how old is it?

4) If silicon is 10^6 times more abundant than gold, how many times more abundant is silicon than gold written as a standard decimal number?

A Characteristic of regime shift/tipping point



A low climate threshold for south Greenland Ice Sheet demise during the Late Pleistocene

Nil Irvali[a,b,1], Eirik V. Galaasen[a,b], Ulysses S. Ninnemann[a,b], Yair Rosenthal[c,d], Andreas Born[a,b], and Helga (Kikki) F. Kleiven[a,b]

[a]Department of Earth Science, University of Bergen, 5007 Bergen, Norway; [b]Bjerknes Centre for Climate Research, 5007 Bergen, Norway; [c]Department of Marine and Coastal Sciences, Rutgers, State University of New Jersey, New Brunswick, NJ 08901; and [d]Department of Earth and Planetary Sciences, Rutgers, State University of New Jersey, New Brunswick, NJ 08901

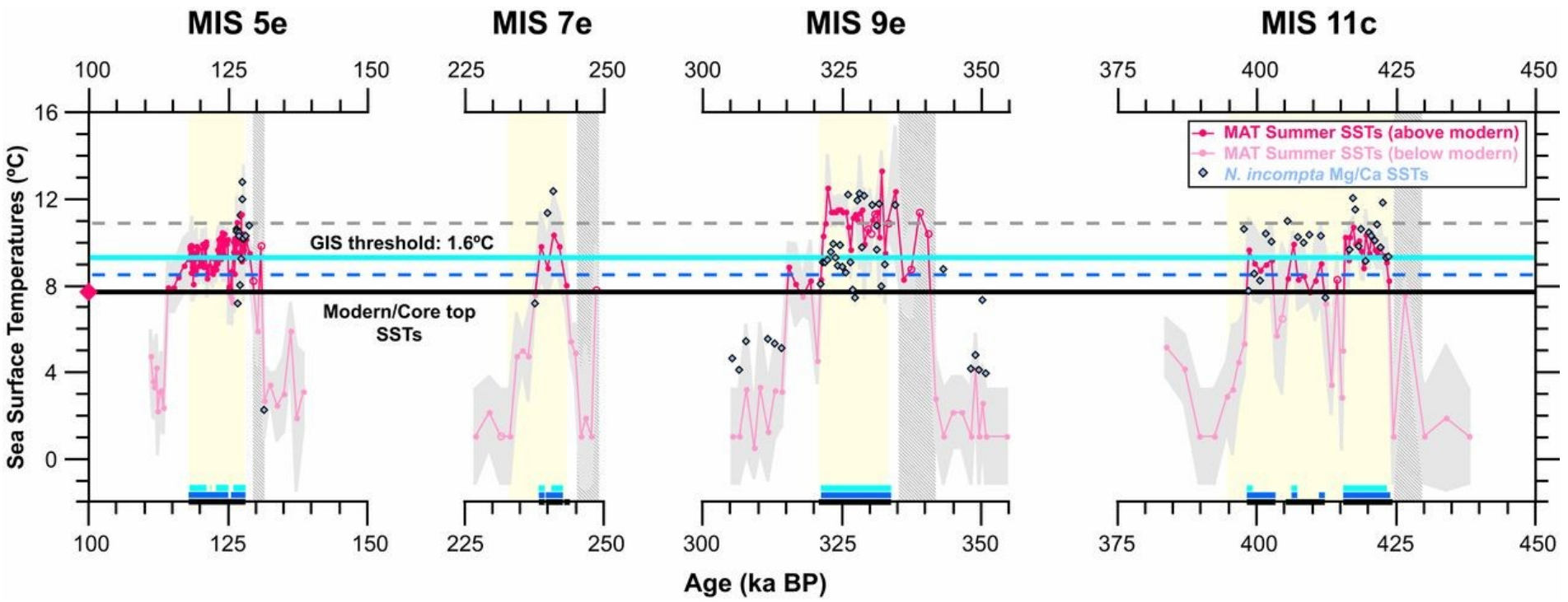
Edited by Gilles Ramstein, Laboratoire des Sciences du Climat et de l'Environnement, Gif-Sur-Yvette, France, and accepted by Editorial Board Member Jean Jouzel November 4, 2019 (received for review July 11, 2019)

The Greenland Ice Sheet (GIS) has been losing mass at an accelerating rate over the recent decades. Models suggest a possible temperature threshold between 0.8 and 3.2 °C, beyond which GIS decline becomes irreversible. The duration of warmth above a given threshold is also a critical determinant for GIS survival, underlining the role of ocean warming, as its inertia prolongs warmth and triggers longer-term feedbacks. The exact point at which these feedbacks are triggered remains equivocal. Late Pleistocene interglacials provide potential case examples for constraining the past response of the GIS to a range of climate states, including conditions warmer than present. However, little is known about the magnitude and duration of warming near Greenland during these periods. Using high-resolution multiproxy surface ocean climate records off southern Greenland, we show that the previous 4 interglacials over the last ~450 ka all reached warmer than present climate conditions and exceeded the modeled temperature threshold for GIS collapse but by different magnitudes and durations. Complete deglaciation of the southern GIS in Marine Isotope Stage 11c (MIS 11c; 394.7 to 424.2 ka) occurred under climates only slightly warmer than present (~0.5 ± 1.6 °C), placing the temperature threshold for major GIS retreat in the lower end of model estimates and within projections for this century.

climatic background associated with and driving these changes remains poorly constrained. Dinocyst-based sea surface temperature (SST) records suggest variable interglacial warmth during Late Pleistocene interglacials close to southern Greenland (5). However, there are no available SST records proximal to southern Greenland that span all recent interglacials, including MIS 9e, which is a particularly interesting interval given strong but incomplete GIS retreat (9).

Here, we provide high-resolution ocean climate (SST) reconstructions from Integrated Ocean Drilling Program (IODP) Site U1305 (57°28.5′ N, 48°31.8′ W; 3,459 m) from the Eirik Drift off southern Greenland, spanning MIS 7e (benthic $\delta^{18}O$ plateau: 232.6 to 243.2 ka), MIS 9e (benthic $\delta^{18}O$ plateau: 320.9 to 333.1 ka), and MIS 11c (benthic $\delta^{18}O$ plateau: 394.7 to 424.2 ka), which we combine with existing MIS 5e (benthic $\delta^{18}O$ plateau: 117.9 to 128.2 ka) records from this location (Core MD03-2664) (10). This site is sensitively situated to monitor subpolar climate near and partially over Greenland (*SI Appendix*, Fig. S1), fluctuations in GIS extent, and iceberg calving events (10, 11). We use benthic foraminifera $\delta^{18}O$ to identify peak interglacial intervals and constrain our age model (*SI Appendix*, Fig. S2). Changes in climate are portrayed using planktonic foraminiferal $\delta^{18}O$, assem-

Greenland Ice Sheet



Irvali et al. (2019) PNAS

Until Wednesday…